

BOME! Bilevel Optimization Made Easy: A Simple First-Order Approach

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Introduction: Bilevel Optimization (BO)

- ▶ **Problem:** Nested optimization structure.

$$\min_{v, \theta} f(v, \theta) \quad \text{s.t.} \quad \theta \in \arg \min_{\theta'} g(v, \theta')$$

- ▶ **Variables:** Outer variable $v \in \mathbb{R}^m$, inner variable $\theta \in \mathbb{R}^n$.
- ▶ **Applications:**
 - ▶ Hyperparameter Optimization (f - val loss; g - train loss)
 - ▶ Meta-Learning
 - ▶ Continual Learning
 - ▶ Reinforcement Learning
- ▶ **Challenge:** The nested nature makes BO notoriously difficult.

Background: Traditional Approaches

1. Hypergradient Descent (Implicit Diff)

- ▶ Assumes a unique inner solution $\theta^*(v) = \arg \min_{\theta} g(v, \theta)$.
- ▶ Requires computing $\nabla_v \theta^*(v)$ via implicit function theorem:

$$\nabla_v \theta^*(v) = -[\nabla_{2,2}g]^{-1} \nabla_{1,2}g$$

- ▶ **Problems:** Involves expensive Hessian manipulations (inverse, CG, Neumann series).

2. Stationary-Seeking Methods

- ▶ Replaces $\arg \min$ with $\nabla_{\theta} g(v, \theta) = 0$.

$$\min_{v, \theta} f(v, \theta) \quad \text{s.t.} \quad \nabla_{\theta} g(v, \theta) = 0$$

- ▶ **Problems:** Only equivalent if g is convex in θ ; unsuitable for non-convex deep learning.

Method: Value-Function Based Reformulation

Key Insight: Reformulate (1) as a single-level constraint problem.

$$\min_{v, \theta} f(v, \theta) \quad \text{s.t.} \quad q(v, \theta) := g(v, \theta) - g^*(v) \leq 0$$

where $g^*(v) = \min_{\theta} g(v, \theta)$ is the **value function**.

Why is this powerful?

- ▶ Removes the nested structure.
- ▶ By Danskin's Theorem, $\nabla_v g^*(v) = \nabla_1 g(v, \theta^*(v))$. We can do it because $\nabla_v g^*(v)$ doesn't depend from $\nabla_v \theta^*(v)$
- ▶ **No implicit derivative $\nabla_v \theta^*$ required!**

This formulation is valid even for non-convex g .

Method: Dynamic Barrier Gradient Descent

We solve the constrained problem using a **dynamic barrier method**. Idea is controlling the decrease of the constraint q , ensuring that q decreases whenever $q > 0$

Core Update

$$(v_{k+1}, \theta_{k+1}) \leftarrow (v_k, \theta_k) - \xi \delta_k$$

where $\delta_k = \arg \min_{\delta} \|\nabla f(v_k, \theta_k) - \delta\|^2$ s.t. $\langle \nabla q_k, \delta_k \rangle \geq \phi_k \geq 0$

Closed-Form Solution

$$\delta_k = \nabla f_k + \lambda_k \nabla q_k, \quad \lambda_k = \max \left(\frac{\phi_k - \langle \nabla f_k, \nabla q_k \rangle}{\|\nabla q_k\|^2}, 0 \right)$$

Choices for ϕ_k : $\phi_k = \eta q_k$ or $\phi_k = \eta \|\nabla q_k\|^2$.

Algorithm: BOME in Practice

Algorithm 1 BOME (Bilevel Optimization Made Easy)

```
1: Input: Initial  $(v_0, \theta_0)$ , step size  $\xi$ , inner steps  $T$ , barrier coeff.  
    $\eta$ .  
2: for  $k = 0, 1, \dots$  do  
3:    $\theta_k^{(0)} = \theta_k$   
4:   for  $t = 0$  to  $T - 1$  do  
5:      $\theta_k^{(t+1)} = \theta_k^{(t)} - \alpha \nabla_{\theta} g(v_k, \theta_k^{(t)})$  ▷ Inner GD  
6:   end for  
7:    $\hat{q}(v, \theta) = g(v, \theta) - g(v, \theta_k^{(T)})$  ▷ Approx. value function  
   (stop-gradient)  
8:   Compute  $\nabla \hat{q}(v_k, \theta_k)$  ▷ No gradient through  $\theta_k^{(T)}$   
9:    $\phi_k = \eta \|\nabla \hat{q}(v_k, \theta_k)\|^2$  (default)  
10:   $(v_{k+1}, \theta_{k+1}) \leftarrow (v_k, \theta_k) - \xi(\nabla f_k + \lambda_k \nabla \hat{q}_k)$   
11: end for
```

$\theta_k^{(T)}$ is treated as a constant when computing $\nabla \hat{q}$ (stop-gradient).

Theoretical Analysis: KKT Conditions

For problem, standard KKT fails because CRCQ (constant rank constraint quantification) doesn't hold ($\nabla q = 0$ at solution).

- ▶ We use a **proximal KKT measure** to analyze convergence:

$$\mathcal{K}(v, \theta) = \underbrace{\min_{\lambda \geq 0} \|\nabla f(v, \theta) + \lambda \nabla q(v, \theta)\|^2}_{\text{Local improvement}} + \underbrace{q(v, \theta)}_{\text{Feasibility}}$$

Proposition 1 (Connection to True Stationarity)

If a sequence satisfies $\nabla f_k + \lambda_k \nabla q_k \rightarrow 0$ and $q_k \rightarrow 0$, its limit point satisfies the KKT conditions of the stationary-seeking formulation.

$\mathcal{K}(v_k, \theta_k) \rightarrow 0; k \rightarrow +\infty$ is our target.

Theoretical Analysis: Unimodal g (Global Optimum)

Assumption 1: PL Inequality

$$\|\nabla_{\theta} g(v, \theta)\|^2 \geq \kappa(g(v, \theta) - g^*(v))$$

Theorem (Convergence Rate for Unimodal g)

Under smoothness and PL assumptions, with $\phi_k = \eta \|\nabla \hat{q}_k\|^2$ and T sufficiently large:

$$\min_{k \leq K} \mathcal{K}(v_k, \theta_k) = O\left(\sqrt{\xi} + \sqrt{\frac{q_0}{\xi K}} + \frac{1}{\xi K} + e^{-bT}\right)$$

where $q_0 = q(v_0, \theta_0)$.

Interpretation:

- ▶ Rate depends on initial inner optimality q_0 .
- ▶ With $\xi = O(K^{-1/2})$, we get $\min \mathcal{K} = O(K^{-1/4} + e^{-bT})$.
- ▶ First non-asymptotic rate for a fully first-order BO method.

Theoretical Analysis: Multimodal g (Local Optima)

Assumption 2: Local PL within attraction basins

- ▶ $\theta^\circ(v, \theta)$ is the "attraction point" of GD from θ on $g(v, \cdot)$.
- ▶ $q^\circ(v, \theta) = g(v, \theta) - g(v, \theta^\circ(v, \theta))$

Theorem (Convergence Rate for Multimodal g)

Under local PL and smoothness, if the algorithm never hits basin boundaries, then:

$$\min_{k \leq K} \mathcal{K}^\circ(v_k, \theta_k) = O\left(\sqrt{\xi} + \sqrt{\frac{1}{\xi K}} + e^{-bT}\right)$$

Interpretation:

- ▶ Rate does **not** improve with small initial q_0° , as the attraction basin may change.
- ▶ With $\xi = O(K^{-1/2})$, we get $\min \mathcal{K}^\circ = O(K^{-1/4} + e^{-bT})$.

Experiments: Hyperparameter Optimization

MNIST:

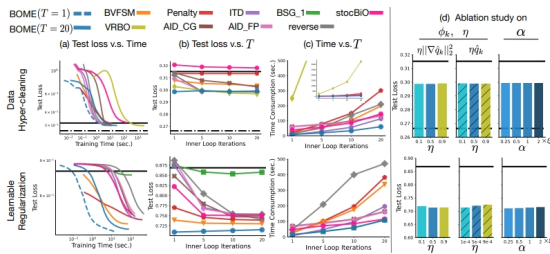


Figure: Top: data hyper-cleaning on MNIST dataset. Bottom: learnable regularization on 20 Newsgroup dataset.

Experiments: Continual Learning

Contextual Transformation Network (CTN):

- ▶ Train on a sequence of tasks (PMNIST, Split CIFAR).
- ▶ Metrics: Accuracy (ACC $\uparrow$), Negative Backward Transfer (NBT $\downarrow$), Forward Transfer (FT $\uparrow$).

Method (PMNIST)	ACC ($\uparrow$)	NBT ($\downarrow$)	FT ($\uparrow$)
CTN (+ITD)	78.40	5.62	84.02
CTN (+BVFSM)	77.78	7.25	85.03
CTN (+BOME)	80.70	4.09	84.79

Table: BOME provides a substantial performance boost over other bilevel optimizers in CL.

Key Observations & Robustness

Why BOME works well:

- ▶ **Faster Learning:** Converges to better solutions.
- ▶ **Robust to Parameters:**
 - ▶ Default $\eta = 0.5$, $T = 10$ works across tasks.
 - ▶ $T = 1$ is often sufficient!
- ▶ **Choices for ϕ_k :** Both $\phi = \eta \|\nabla \hat{q}\|^2$ and $\phi = \eta \hat{q}$ work similarly with proper scaling.
- ▶ **vs. BVFSM:** BOME is faster, more accurate, and has fewer sensitive hyperparameters.

Conclusion

Summary

BOME is a simple and efficient fully first-order method for bilevel optimization.

- ▶ Built on a **value-function reformulation** and a **dynamic barrier method**.
- ▶ Requires **no Hessian computation** or implicit differentiation.
- ▶ First method with a **non-asymptotic convergence guarantee** for general non-convex BO.

Takeaways

- ▶ BOME is practical and scalable for large-scale deep learning.
- ▶ Empirically outperforms SOTA methods in both speed and final performance.
- ▶ Robust to hyperparameters (η, T) .